HARDWARE DESIGN REVIEW

LAN7800 | USB 3.1 Gen 1 to Gigabit Ethernet Adapter | Rev A

DOCUMENT	REVISION	RELEASE CLASSIFICATION	PROJECT
LAN7800-HDR-001	1.1	Production Approved — design-owner release	LAN7800 Rev A

1. Design review summary

Rev A integrates a Microchip LAN7800 USB-to-Gigabit Ethernet controller, SuperSpeed USB Type-B interface, magnetics-equipped RJ45 connection, buck-regulated power distribution, high-speed ESD protection and a 25 MHz reference clock. The four-layer Altium PCB applies interface-specific routing rules and a controlled-impedance layer stack.

The issued manufacturing data comprises editable schematic and PCB source, a fabrication drawing, Gerber X2 and NC Drill outputs, a controlled BOM and component placement coordinates. The final Altium design-rule report records zero warnings and zero violations.

2. Design deliverables and configuration

WORK PRODUCT	CONFIGURATION / LOCATION	REVIEW RECORD
Altium source	Hardware/Altium/ — PrjPcb, two SchDoc sheets, PcbDoc, PcbLib	Configuration identified
Schematic release	Docs/LAN7800_USB_Ethernet_RevA_Schematic.pdf	Two functional sheets
Fabrication drawing	Manufacturing/LAN7800_USB_Ethernet_RevA_Fabrication.PDF	Outline, stack-up and drill table
Fabrication outputs	Manufacturing/GerberX2/ and Manufacturing/NC Drill/	Four copper layers, masks, paste, drill
BOM / component placement	Manufacturing/BoM/ and Manufacturing/CPL/	22 BOM groups; 48 positions

3. Requirements and design checks

DISCIPLINE	DESIGN REVIEW RECORD	DISPOSITION
USB interface	SuperSpeed and USB 2.0 signal paths; 90 Ω differential routing target.	Documented
Ethernet MDI	RJ45 magnetics; 100 Ω differential routing target and pair-length rules.	Documented
Power delivery	TLV62568 buck supply, power distribution, decoupling and protection.	Documented
Clock and resets	25 MHz reference crystal and LAN7800 control connections.	Documented
Signal integrity	Dedicated differential-pair classes, impedance and matched-length constraints.	DRC checked
Fabrication	Four-layer construction, board profile, plated/non-plated drills, solder mask and paste.	Release data available

4. Detailed technical review

AREA	DESIGN DECISION / REVIEW NOTE
High-speed USB	Maintain the specified 90 Ω differential geometry, controlled pair spacing, closely coupled P/N routes and uninterrupted adjacent reference plane. Verify design rule profiles correspond to the issued stack-up.
Gigabit Ethernet	Maintain 100 Ω differential MDI routing; review magnetics centre-tap network, polarity and pin assignment against the LAN7800 and RJ45 specifications.

Power and thermal	Review TLV62568 power-stage loop geometry, inductor current margin, decoupling locations, return currents and output ripple allocation at the controller supply pins.
ESD / EMC	ESD7008 protection at USB interface; ensure discharge loops and connector-to-ground paths remain short and free of unnecessary stubs.
Oscillator	Confirm 25 MHz crystal parameters, load-capacitance assumptions, trace length, local ground and noise separation around oscillator pins.
DFM / assembly	Check solder-mask apertures, paste requirements, exposed pads, polarity marks, connector fit, pick-and-place origin, rotations and reflow compatibility.
DFT / inspection	Provide accessible rails and reset test points; define visual/continuity checks and functional USB/Ethernet commissioning sequence for first article.

5. PCB design-rule verification

The Rev A fabrication drawing specifies a 70.00 × 34.00 mm PCB, nominal four-layer IS400/FR-4 construction, USB differential target 90 Ω ±10%, and Gigabit Ethernet differential target 100 Ω ±10%. Nominal stack legend and finished-thickness annotation differ by 0.02 mm (1.53 / 1.55 mm), within typical manufacturing stack tolerances pending the fabricator's controlled stack specification.

The published Altium DRC report, dated 11 August 2026, records 0 warnings and 0 rule violations, including clearance, un-routed nets, hole rules and matched-length constraints. DRC records refer to the configured active rules and must be preserved with the issued revision.

6. Configuration and engineering follow-up

ID	ENGINEERING REVIEW NOTE	RECOMMENDED DOCUMENT CONTROL ACTION
ER-01	C5 is identified as "C0603" in the existing schematic, with no electrical capacitance stated.	Record the selected capacitance on schematic and controlled BOM to maintain agreement.
ER-02	Some component order codes were chosen during BOM preparation.	Retain approved sourcing records with voltage, tolerance, temperature and package requirements.
ER-03	22 µF MLCC groups include compact package selections.	Record effective-capacitance assessment under operating DC bias.
ER-04	CPL includes 48 unique top-layer placements.	Maintain assembly origin/rotation convention and first-article placement checks.

7. Approval and release record

Design-owner release classification: PRODUCTION APPROVED. This report identifies the controlled design outputs and the documented engineering review points for Rev A. Manufacturing planning and configuration management shall use the revision-matched schematic, PCB, BOM, CPL and fabrication outputs.

RELEASE ROLE	RECORD	DATE
Document prepared	CD Designs — Engineering	10 Oct 2026
Design-owner approval	Approval classification specified by design owner	10 Oct 2026

Repository: https://github.com/CDonohoe-Designs/USB3-Gigabit-Ethernet-Adapter-Portfolio